

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCVC405 - HYDRAULICS - II

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

- 1 Velocity in an open channel is maximum at:
a) Top surface b) Near bottom c) Below the surface d) Side wall
- 2 A channel has a hydraulic radius of 0.5 m and slope of 0.001. If Chezy's $C = 50$, velocity is:
a) 1 m/s b) 2.5 m/s c) 3.5 m/s d) 4 m/s
- 3 Hydraulically efficient section minimizes:
a) Velocity b) Flow area c) Wetted perimeter d) Flow depth
- 4 For a most economical rectangular channel section, depth is 2 m. What is bottom width?
a) 2 m b) 3 m c) 4 m d) 5 m
- 5 The hydraulic radius is defined as:
a) Wetted perimeter / Area b) Area / Wetted perimeter c) Depth / Width
d) Width / Depth
- 6 Specific force is used in:
a) Flow resistance b) Uniform flow c) Hydraulic jump analysis d) Turbine design
- 7 A gradually varied flow is one in which:
a) Flow depth changes abruptly b) Flow depth remains constant c) Flow depth changes gradually along the length d) Flow is uniform
- 8 A hydraulic jump is an example of:
a) Gradually varied flow b) Supercritical flow c) Rapidly varied flow
d) Uniform flow
- 9 In subcritical flow, the Froude number is:
a) < 1 b) $= 1$ c) > 1 d) $= 0$
- 10 The momentum of water jet striking a flat plate perpendicularly is converted into:
a) Only kinetic energy b) Heat and sound c) Force on the plate d) Torque
- 11 Which turbine is used for high heads?
a) Francis b) Kaplan c) Pelton d) Axial flow

- 12 Pump cavitation is caused due to:
 a) Low flow velocity b) High NPSH c) Vapor formation at low pressure
 d) Sudden expansion
- Q2. Solve the following.
- A) Explain in detail momentum equation applied to open channel flow. 6
- B) Determine the height of a rectangular weir of length 6 m to be built across a rectangular channel. The maximum depth of water on the upstream side of the weir is 1.8 m and discharge is 2000 liters/s. Take $C_d = 0.6$ and neglect end contraction. 6
- Q3. Solve the following.
- A) What do you understand by 'Economical section' or 'Hydraulically most efficient section'? Derive condition for hydraulically most efficient rectangular section. 6
- B) What is specific energy? Describe Specific energy curve with neat diagram. 6
- Q4. Solve Any Two of the following.
- A) Derive the following dynamic equation for gradually varied flow. 6
- $$\frac{dy}{dx} = \frac{S_0 - S_f}{1 - \frac{Q^2 T}{g A^3}}$$
- B) Derive an expression for impact of jet of water on stationary flat plate. 6
- C) A sluice gate discharges water into a horizontal rectangular channel with velocity of 5 m/s and depth of flow 0.35 m, width of channel is 7 m. Determine whether hydraulic jump will occur and if so, determine its height and loss of energy per newton of water. <https://www.batuonline.com> 6
- Q5. Solve Any Two of the following.
- A) Distinguish between impulse turbine and reaction turbine. 6
- B) Explain the following terms: 6
 (i) Hydraulic efficiency; (ii) Mechanical efficiency; (iii) Overall efficiency
- C) Enlist types of turbines. Explain with neat sketch Pelton wheel turbine. 6
- Q6. Solve Any Two of the following.
- A) Explain in detail component parts of centrifugal pumps with neat diagram. 6
- B) The internal and external diameters of the impeller of a centrifugal pump are 200 mm and 400 mm respectively. The pump is running at 1200 rpm. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially, and the velocity of flow is constant. Determine the work done by the impeller per unit weight of water. 6
- C) Define priming. Why it is necessary for a centrifugal pump? 6

*** End ***